

Interactions

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What is an interaction?

Definition

An **interaction** is an additional term in a regression model that allows the **slope** of one variable to depend on the **value** of another.

Two categorical variables

A categorical and a quantitative variable

Two quantitative variables



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- In other words, does the **effect** of one variable (i.e., its slope coefficient) depend on the **value** of another?
- This is what **interactions** let us model



The algebra of interactions

The idea is to add a term that is the product of the two variables:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1(\text{male}) + \hat{\beta}_2(\text{tenured}) + \hat{\beta}_3(\text{male})(\text{tenured})$$



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For female professors, $\text{male} = 0$, so the β_1 and β_3 terms cancel out:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_2(\text{tenured})$$

For male professors, $\text{male} = 1$, so we get both a different intercept and a different slope for tenured :

$$\hat{Y} = (\hat{\beta}_0 + \hat{\beta}_1) + (\hat{\beta}_2 + \hat{\beta}_3)(\text{tenured})$$

```
model1 <- lm(eval ~ gender * tenure, data=profs)
summary(model1)
```

Call:

```
lm(formula = eval ~ gender * tenure, data = profs)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.89028	-0.36000	0.00972	0.40972	1.00972

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.86000	0.07585	50.890	< 2e-16 ***
gendermale	0.53615	0.10623	5.047	6.48e-07 ***
tenureyes	0.05517	0.08796	0.627	0.530813
gendermale:tenureyes	-0.46105	0.12083	-3.816	0.000154 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5363 on 459 degrees of freedom

Multiple R-squared: 0.07173, Adjusted R-squared: 0.06567

F-statistic: 11.82 on 3 and 459 DF, p-value: 1.795e-07



Main effects and interaction effects

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- In other words, the main effect `gendermale` represents the predicted advantage for men in student evaluation scores, among professors without tenure



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- In a model with an interaction term X_1X_2 , you **must** also keep the **main effects**: the variables that are being interacted together
- The main effect of X_1 represents the predicted increase in Y for a 1-unit change in X_1 , holding X_2 constant **at zero**
- In other words, the main effect `gendermale` represents the predicted advantage for men in student evaluation scores, among professors without tenure
- You can also include other variables in the model that are not being interacted

Two categorical variables

A categorical and a quantitative variable

Two quantitative variables



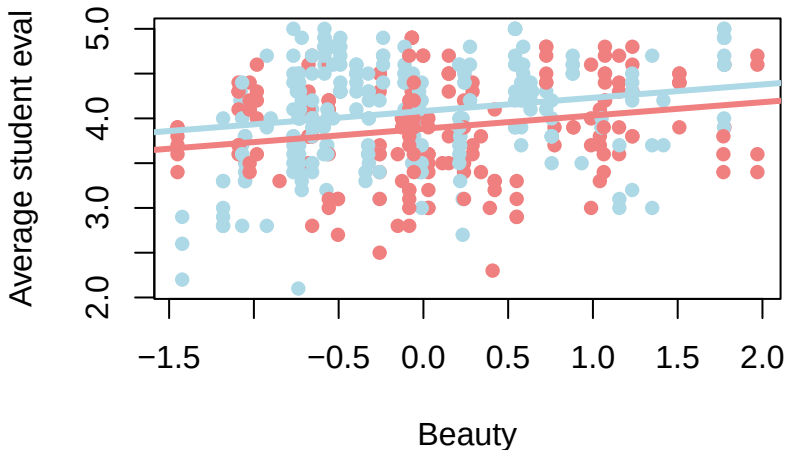
Does beauty matter more for men, or for women?

- We found that for the same level of attractiveness, male professors tend to get higher evaluation scores than female professors
- But what if the **effect** of beauty depend on gender?



Does beauty matter more for men, or for women?

Another way to think about it—what if these regression lines didn't have to be parallel?



```
model2 <- lm(eval ~ beauty * gender, data=profs)
summary(model2)
```

Call:

```
lm(formula = eval ~ beauty * gender, data = profs)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.83820	-0.37387	0.04551	0.39876	1.06764

Coefficients:

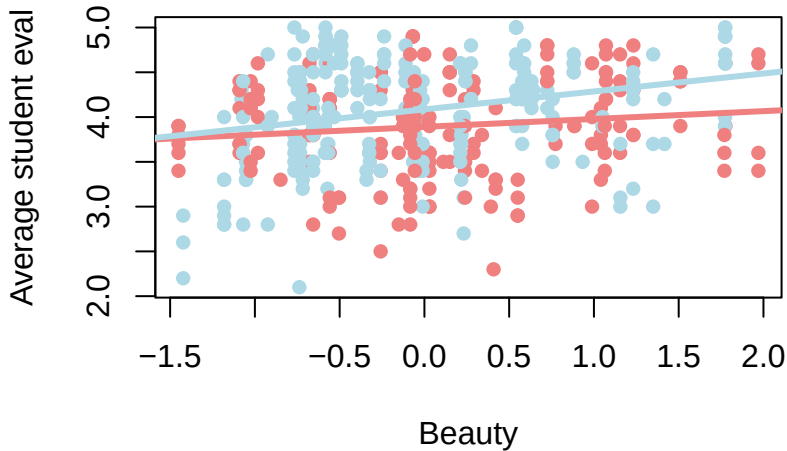
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.89085	0.03878	100.337	< 2e-16 ***
beauty	0.08762	0.04706	1.862	0.063294 .
gendermale	0.19510	0.05089	3.834	0.000144 ***
beauty:gendermale	0.11266	0.06398	1.761	0.078910 .

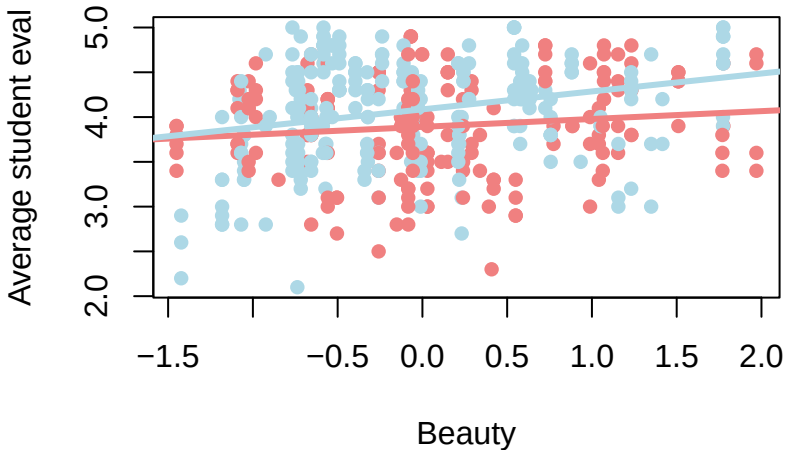
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5361 on 459 degrees of freedom

Multiple R-squared: 0.07256, Adjusted R-squared: 0.0665

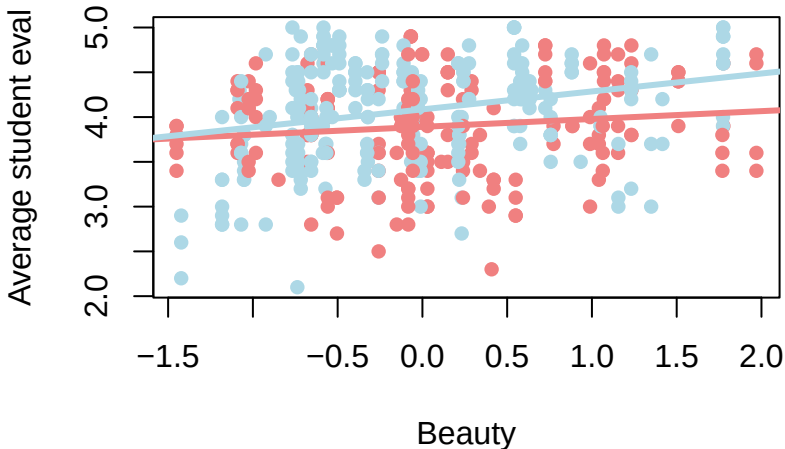
F-statistic: 11.97 on 3 and 459 DF, p-value: 1.47e-07





Two takeaways:

- Beauty seems to matter more for men than for women!



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- Beauty seems to matter more for men than for women!
- The gender gap is largest for good-looking professors

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Two quantitative variables

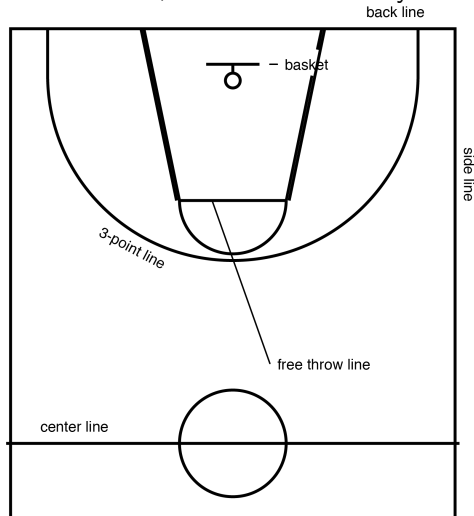


Basketball-Reference.com provides detailed data on NBA teams and players. We'll look at team data for 4 seasons ending in 2016; each of these metrics is the average across the season:

- **PTS**: Total points
- **PCT3P**: Percentage of 3-point shots made
- **N3PA**: Number of 3-point shots attempted

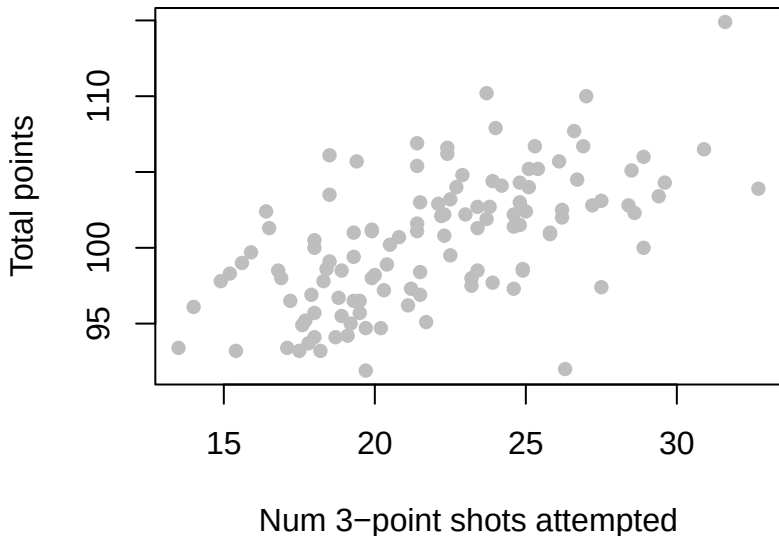
There are $30 \text{ NBA teams} \times 4 \text{ seasons} = 120$ cases in this file.

In basketball, there are three ways to score:



- **1 point** for free throws made after a foul by the other team
- **2 points** for shots made inside the 3-point line
- **3 points** for shots made outside the 3-point line

The more 3-pointers you attempt, the more you tend to score:



```
model1 <- lm(PTS ~ N3PA, data=nba)
summary(model1)
```

```
Call:
lm(formula = PTS ~ N3PA, data = nba)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-11.2454	-2.5114	0.0549	2.2252	8.6405

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	86.19204	1.77464	48.569	< 2e-16 ***
N3PA	0.64842	0.07935	8.171	3.89e-13 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 3.496 on 118 degrees of freedom
```

```
Multiple R-squared:  0.3614, Adjusted R-squared:  0.356
```

```
F-statistic: 66.77 on 1 and 118 DF,  p-value: 3.889e-13
```



Can we do better?

- $R^2 = 36\%$, so we can explain 36% of the variance in total points based only on knowing the number of 3-point attempts.

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- This means that **most** of the variance (64%) in total points is **not** explained by the number of 3-point attempts.

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- $R^2 = 36\%$, so we can explain 36% of the variance in total points based only on knowing the number of 3-point attempts.
- This means that **most** of the variance (64%) in total points is **not** explained by the number of 3-point attempts.
- Let's add another variable to our model — why might 3-point percentage be useful as another predictor?



Can we do better?

```
model2 <- lm(PTS ~ N3PA + PCT3P, data=nba)
summary(model2)
```

```
Call:
lm(formula = PTS ~ N3PA + PCT3P, data = nba)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.3487	-2.1392	-0.0791	1.8691	9.1904

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	62.00493	5.61396	11.045	< 2e-16 ***
N3PA	0.56467	0.07587	7.442	1.82e-11 ***
PCT3P	0.73415	0.16292	4.506	1.57e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.241 on 117 degrees of freedom

Multiple R-squared: 0.4558, Adjusted R-squared: 0.4465

F-statistic: 49 on 2 and 117 DF, p-value: 3.478e-16



Can we do even better?

It would make sense that the **impact** of the number of 3-pointers taken on total points would **depend on** how well the team shoots the 3!

Can we do even better?



It would make sense that the **impact** of the number of 3-pointers taken on total points would **depend on** how well the team shoots the 3!

This sounds like an interaction — let's make a model with an interaction between the two predictors!

```
model3 <- lm(PTS ~ N3PA * PCT3P, data=nba)
summary(model3)
```

```
Call:
lm(formula = PTS ~ N3PA * PCT3P, data = nba)
```

```
Residuals:
```

	Min	1Q	Median	3Q	Max
	-7.2629	-2.2757	0.1148	1.9698	9.3756

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	122.84903	30.58937	4.016	0.000105 ***
N3PA	-2.11904	1.32903	-1.594	0.113561
PCT3P	-0.98410	0.86465	-1.138	0.257400
N3PA:PCT3P	0.07561	0.03739	2.023	0.045423 *

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 3.199 on 116 degrees of freedom
Multiple R-squared:  0.4743, Adjusted R-squared:  0.4608
F-statistic: 34.89 on 3 and 116 DF,  p-value: 3.798e-16
```

Model 3 corresponds to the regression equation

$$\widehat{\text{PTS}} = 122.85 - 2.12 \cdot \text{N3PA} - 0.98 \cdot \text{PCT3P} + 0.08 \cdot \text{N3PA} \cdot \text{PCT3P}.$$

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We interpret the coefficients as follows:

- **Intercept** (122.85) is our prediction of total points when $\text{N3PA} = \text{PCT3P} = 0$. (Meaningless in this context!)

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- **N3PA** (-2.12) is the predicted increase in total points for each additional 3-pointer taken, when $\text{PCT3P} = 0$.

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- **PCT3P** (-0.98) is the predicted increase in total points for each additional percentage point of 3-point shooting accuracy, when $\text{N3PA} = 0$.

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- **PCT3P** (-0.98) is the predicted increase in total points for each additional percentage point of 3-point shooting accuracy, when $\text{N3PA} = 0$.
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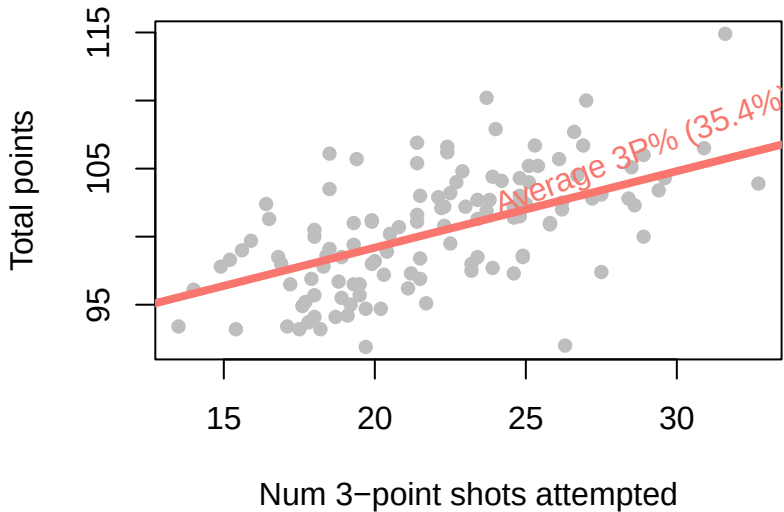
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 - the increase in the *slope coefficient* for N3PA for each 1-unit increase of PCT3P.

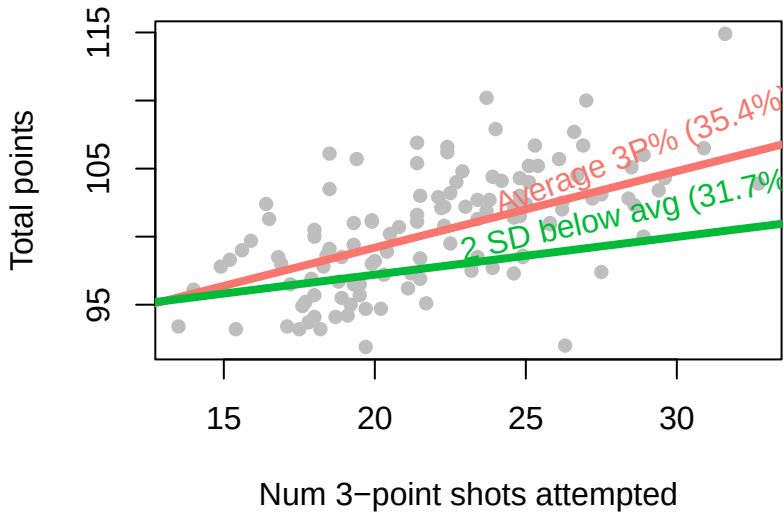
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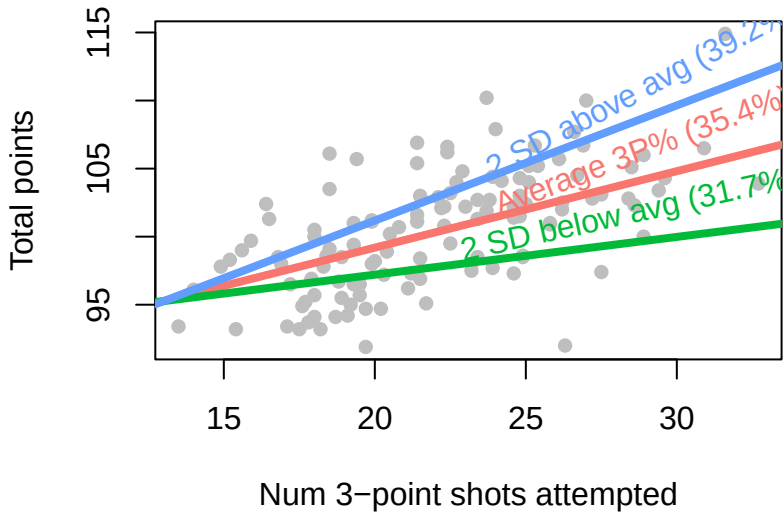
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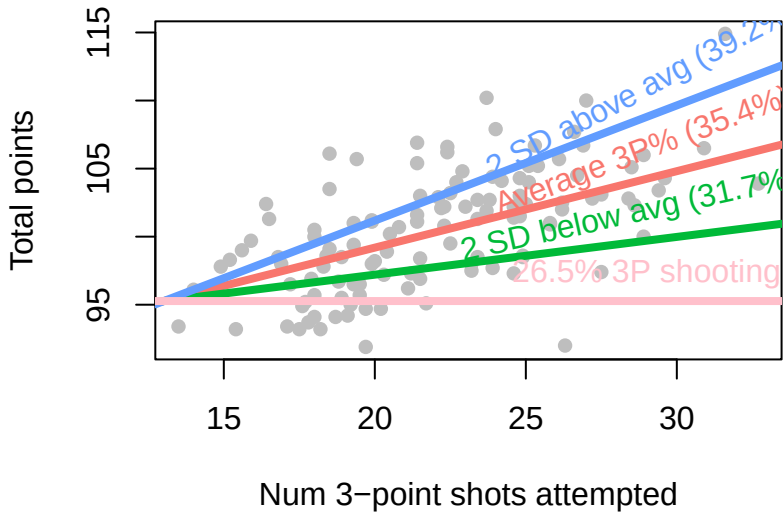


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- How many points per game do you predict for a team that shoots 3-pointers at the NBA average rate (35.4) and that takes 30 3-pointers per game?

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- How many points per game do you predict for a team that shoots 3-pointers at the NBA average rate (35.4) and that takes 30 3-pointers per game?
- How bad would a team have to shoot the 3 before taking 3-point shots start to have a negative impact on total points?





When should you use interactions in a model?

- Choose interactions by thinking about what you are trying to model: if you suspect that the impact of one variable depends on the value of another, try an interaction term between them!
- But remember: Interactions make a model more complex to analyze and explain, so unless you have a good **substantive** reason it's usually only worth doing so when you get a substantial bump in R^2 /decrease in residual standard error by including the interaction